

Mastering CSS Grid: A Complete Guide

Understanding CSS Grid

CSS Grid Layout is a powerful layout system designed to create complex and responsive web designs with ease. It allows for precise control over rows and columns, making it a superior alternative to older layout methods like float and flexbox for grid-based designs.

Key CSS Grid Properties

1. Parent (Grid Container) Properties

display: grid | inline-grid

Defines a grid container. Example:

```
.container {  
  display: grid;  
}
```

grid-template-columns & grid-template-rows

Defines the number and size of columns and rows. Example:

```
.container {  
  display: grid;  
  grid-template-columns: 200px 1fr 2fr;  
  grid-template-rows: 100px auto;  
}
```

grid-gap, row-gap, column-gap

Sets spacing between grid items. Example:

```
.container {  
  display: grid;  
  grid-gap: 20px;
```

```
}
```

justify-content

Aligns the grid within the container along the main axis. Values:

- start | end | center | space-between | space-around | space-evenly

Example:

```
.container {  
  justify-content: center;  
}
```

align-content

Aligns grid rows within the container. Values:

- start | end | center | space-between | space-around | stretch

Example:

```
.container {  
  align-content: stretch;  
}
```

justify-items

Aligns items inside grid cells along the row axis. Example:

```
.container {  
  justify-items: center;  
}
```

align-items

Aligns items inside grid cells along the column axis. Example:

```
.container {  
  align-items: start;
```

```
}
```

grid-auto-flow: row | column | dense

Controls how items are placed automatically. Example:

```
.container {  
  grid-auto-flow: column;  
}
```

2. Child (Grid Item) Properties

grid-column & grid-row

Defines how many columns/rows an item should span. Example:

```
.item {  
  grid-column: 1 / span 2;  
  grid-row: 2 / 3;  
}
```

justify-self & align-self

Aligns an individual item inside its cell. Example:

```
.item {  
  justify-self: end;  
  align-self: center;  
}
```

grid-area

Assigns an item to a named area. Example:

```
.item {  
  grid-area: header;
```

}